# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

# BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Biology (Objective) Group - I

**Time Allowed:** 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5467

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. World Heart Day is celebrated on:

(A) 28 November
(B) 28 October
(C) 28 September
(D) 28 December

ورلڈ ہارٹ ڈے منایا جاتا ہے:

**(A)** 28 نومبر
**(B)** 28 اکتوبر
**(C)** 28 ستمبر
**(D)** 28 دسمبر

Q2. Average volume of blood in adult human body is about:

(A) 3 liters
(B) 9 liters
(C) 7 liters
(D) 5 liters

بالغ انسان کے جسم میں خون کا اوسط حجم تقریباً ہے:

**(A)** 3 لیٹر
**(B)** 9 لیٹر
**(C)** 7 لیٹر
**(D)** 5 لیٹر

Q3. Which acid converts the inactive enzyme pepsinogen into its active form?

(A) Nitric acid
(B) Sulphuric acid
(C) Hydrochloric acid
(D) Acetic acid

کون سا ایسڈ غیر فعال انزائم پیپسینوجن کو اس کی فعال حالت میں تبدیل کرتا ہے؟

**(A)** نائٹرک ایسڈ
**(B)** سلفیورک ایسڈ
**(C)** ہائیڈروکلورک ایسڈ
**(D)** ایسیٹک ایسڈ

Q4. Bacteria living in colon produce:

(A) Vitamin A
(B) Vitamin K
(C) Vitamin C
(D) Vitamin D

کولون میں رہنے والے بیکٹیریا پیدا کرتے ہیں:

**(A)** وٹامن A
**(B)** وٹامن K
**(C)** وٹامن C
**(D)** وٹامن D

Q5. Stomata cover the leaf surface: _____

(A) 1-2%
(B) 3-4%
(C) 2-3%
(D) 4-5%

سٹومیٹا پتے کی سطح کا _____ ڈھانپتے ہیں:

Q6. The scientist which gave the concept of metabolism first is:

(A) Emil Fischer
(B) Ibn-e-Nafees
(C) Jabir Bin Hayyan
(D) Winhelm Kuhne

وہ سائنسدان جس نے میٹابولزم کا تصور سب سے پہلے دیا:

**(A)** ایمل فشر
**(B)** ابن نفیس
**(C)** جابر بن حیان
**(D)** ون ہیلم کوہنے

Q7. Meiosis was discovered by:

(A) Pasteur
(B) Golgi
(C) Walther Fleming
(D) Oscar Hertwig

می اوسس کس نے دریافت کیا:

**(A)** پاسچر نے
**(B)** گالجی نے
**(C)** والٹر فلیمنگ نے
**(D)** آسکر ہرٹوگ نے

Q8. Process of formation of new tumors is called as:

(A) Regeneration
(B) Crossing over
(C) Metastasis
(D) Synapsis

نئے ٹیومر بننے کا عمل کہلاتا ہے:

**(A)** ری جنریشن
**(B)** کراسنگ اوور
**(C)** میٹا سٹیسس
**(D)** سائنیپسس

Q9. Ribosomes are constructed in: _____

(A) Nucleolus
(B) Plastids
(C) Endoplasmic reticulum
(D) Cell wall

رائبوسومز _____ میں تیار کیے جاتے ہیں۔

**(A)** نیوکلیولس
**(B)** پلاسٹڈز
**(C)** اینڈوپلازمک ریٹیکولم
**(D)** سیل وال

Q10. The basic unit of classification is:

(A) Family
(B) Species
(C) Class
(D) Genus

کلاسیفیکیشن کی بنیادی اکائی ہے:

**(A)** فیملی
**(B)** سپیشیز
**(C)** کلاس
**(D)** جینس

Q11. The first step in solving a biological problem is:

(A) Deduction
(B) Hypothesis
(C) Experiments
(D) Observations

بائیولوجیکل پرابلم کو حل کرنے کا پہلا مرحلہ ہے:

**(A)** ڈیڈکشن
**(B)** ہائپوتھیسس
**(C)** تجربات
**(D)** مشاہدات

Q12. Similar cells organized into groups and performing same functions are called as:

(A) Tissue
(B) Organelle
(C) Organ
(D) Organ system

ایک جیسے سیلز جو گروپ کی شکل میں ترتیب پائے ہوں اور ایک ہی کام کرتے ہوں کہلاتے ہیں:

**(A)** ٹشو
**(B)** آرگینیل
**(C)** آرگن
**(D)** آرگن سسٹم

## Biology (Subjective) Group - I

**Time Allowed:** 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5467

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define immunology.

(ii) Define population and give example.

(iii) Write scientific names of frog and mustard plant.

(iv) Write quantitative and qualitative observations.

(v) What is meant by summarization of results?

(vi) Differentiate between theory and law.

(vii) Differentiate between chromosomes and chromatin.

(viii) Write the functions of ribosomes.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Write aims of classification.

(ii) Differentiate between prions and viroids.

(iii) Give scientific names of onion and see star

(iv) What is phragmoplast?

(v) Give role of mitosis in development and growth.

(vi) What happens during G1 phase of inter phase?

(vii) Give names of sub-units of ATP

(viii) What is Z-scheme and why is it called so?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Describe lock and key model of enzyme action.

(ii) Define cofactors and give example.

(iii) What is meant by metabolism?

(iv) What is meant by churning of food?

(v) Define villi. What is their function?

(vi) What is meant by transpiration pull?

(vii) What is the difference between source and sink?

(viii) Write the symptoms of Dengue fever.

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Define cellular organization. Explain its three types.

Q5(b). (Marks 04) Write a note on cell membrane.

Q6(a). (Marks 05) Define binomial nomenclature. Describe its significance.

Q6(b). (Marks 04) Write a detailed note on oxidation-reduction reactions.

Q7(a). (Marks 05) Write any four functions of liver besides digestion.

Q7(b). (Marks 04) Write detailed note on white blood cells.

## GROUP – II

### Biology (Objective) Group - II

**Time Allowed:** 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5466

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Meiosis was discovered in year:

(A) 1976
(B) 1876
(C) 1786
(D) 1796

می اوسس کی دریافت کب ہوئی؟

Q2. In which phase cell duplicates its chromosomes?

(A) S-Phase
(B) G1-Phase
(C) G2-Phase
(D) G0-Phase

سیل کس فیز میں اپنے کروموسومز کو ڈپلیکیٹ کرتا ہے؟

**(A)** فیز-S
**(B)** فیز-G1
**(C)** فیز-G2
**(D)** فیز-G0

Q3. Inner membrane of mitochondria forms many infoldings called:

(A) Cisternae
(B) Stroma
(C) Grana
(D) Cristae

مائٹوکانڈریا کی اندرونی ممبرین اندر کی طرف بہت سی بناتی ہے:

**(A)** سسٹرنی
**(B)** سٹروما
**(C)** گرانا
**(D)** کرسٹی

Q4. Prokaryotic organisms are included in kingdom:

(A) Protista
(B) Monera
(C) Fungi
(D) Plantae

پروکیریاٹک جانداروں کو کس کنگڈم میں شامل کیا جاتا ہے؟

**(A)** پروٹسٹا
**(B)** مونیرا
**(C)** فنجائی
**(D)** پلانٹی

Q5. Ratio may be expressed by:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) :
(D) ?

تناسب کو کس علامت سے ظاہر کیا جاتا ہے؟

Q6. This division of biology deals with the study of plants:

(A) Zoology
(B) Microbiology
(C) Botany
(D) Morphology

بائیولوجی کی اس ڈویژن کا تعلق پودوں کے مطالعے سے ہے:

**(A)** زوالوجی
**(B)** مائیکروبائیولوجی
**(C)** باٹنی
**(D)** مارفولوجی

Q7. Which is responsible for blood clot?

(A) Monocytes
(B) Neutrophils
(C) Lymphocytes
(D) Basophils

خون کے جمنے کا ذمہ دار ہے؟

**(A)** مونوسائٹس
**(B)** نیوٹروفلز
**(C)** لمفوسائٹس
**(D)** بیسوفلز

Q8. Valves to prevent the backflow of blood found in:

(A) Arteries
(B) Veins
(C) Capillaries
(D) All these

خون کے واپس بہاؤ کو روکنے کے لیے والوز کس میں پائے جاتے ہیں؟

**(A)** آرٹریز
**(B)** وینز
**(C)** کیپلریز
**(D)** یہ تمام

Q9. One gram of proteins contains _____ energy.

(A) 4 k cal
(B) 9 k cal
(C) 70 k cal
(D) 100 k cal

1 گرام پروٹین سے _____ توانائی حاصل ہوتی ہے:

Q10. Which mineral acts as co-factor for enzyme?

(A) Sodium
(B) Potassium
(C) Calcium
(D) Chloride

کون سا منرل اینزائم کے طور پر کام کرتا ہے؟

**(A)** سوڈیم
**(B)** پوٹاشیم
**(C)** کیلشیم
**(D)** کلورائیڈ

Q11. On breaking of one phosphate bond releases energy:

(A) 7.3 cal
(B) 730 k cal
(C) 7300 k cal
(D) 7.3 k cal

ATP کے ایک فاسفیٹ بانڈ کے ٹوٹنے سے انرجی خارج ہوتی ہے:

Q12. The term enzyme was first used by:

(A) Emil Fischer
(B) Danial Koshland
(C) Winhelm Kuhne
(D) Karl Lohmann

انزائم کی اصطلاح پہلی بار کس نے استعمال کی؟

**(A)** ایمل فشر
**(B)** ڈینئل کوشلینڈ
**(C)** ون ہیلم کوہنے
**(D)** کارل لوہمین

## Biology (Subjective) Group - II

**Time Allowed:** 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5466

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) What is meant by fossils?

(ii) Define organ and give an example.

(iii) Write two uses of mustard plant.

(iv) What is the role of biological method in biology?

(v) Differentiate between quantitative and qualitative observations.

(vi) What is meant by reporting the results?

(vii) What is the difference between primary cell wall and secondary cell wall?

(viii) Define filtration.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Write the importance of biodiversity.

(ii) Define taxon.

(iii) Define binomial nomenclature.

(iv) What are germ line cells?

(v) Define mitosis and meiosis.

(vi) What are the functions of centrosomes?

(vii) Write the importance of ATP.

(viii) When and who discovered ATP?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) What is meant by activation energy?

(ii) Write the difference between anabolism and catabolism.

(iii) What are the benefits of biocatalysts?

(iv) Write two functions of oral cavity

(v) What do you know about appendix?

(vi) What is "Cohesion Tension Theory"?

(vii) Write the functions of white blood cells.

(viii) Write preventive measures of dengue fever.

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 05) Write a detailed note on molecular level and tissue level.

Q5(b). (Marks 04) Differentiate between prokaryotic and eukaryotic cells.

Q6(a). (Marks 05) Write characteristics of kingdom fungi and kingdom plantae.

Q6(b). (Marks 04) Write a note on active transport.

Q7(a). (Marks 05) Write a note on carbohydrates and proteins.

Q7(b). (Marks 04) Write a note on transport of food in plants.